

SAQUIB FARHAN

Mohammadpur, Dhaka, Bangladesh | +880 1835 531404 | saquibfarhan23@gmail.com
linkedin.com/in/farhan-saquib-8b4350301

SUMMARY

Electrical and Electronic Engineering graduate from the Military Institute of Science and Technology (MIST) with a CGPA of 3.92/4.00, ranked 1st in the Electronics major. Strong research background with a Q1 IEEE journal publication and multiple IEEE conference papers. Passionate about undergraduate teaching, research, and student mentorship.

EDUCATION

B.Sc. in Electrical and Electronic Engineering	2026
Military Institute of Science & Technology (MIST), Dhaka	CGPA: 3.92/4.00
Ranked 1st in the Electronics major.	
Higher Secondary Certificate (HSC)	2021
Dhaka Residential Model College, Dhaka	GPA: 5.00/5.00
Secondary School Certificate (SSC)	2019
Mohammadpur Preparatory School & College, Dhaka	GPA: 5.00/5.00

RESEARCH INTERESTS

Semiconductor Devices and Nanoelectronics | TCAD-Based Device Modeling and Simulation | VLSI and Low-Power Digital IC Design | Photonic and Optical Biosensors | Artificial Intelligence

PUBLICATIONS

Journal

1. S. Rahman*, **S. Farhan***, R.R. Sharma, K.N. Faisal, "HatSeqNet: A Sequence-based Deep Learning Model for Cross-Subject Driver Drowsiness Detection using Single-Channel EEG Signal," *IEEE Sensors Journal* (Q1), 2026. *S. Rahman and S. Farhan contributed equally.

Conference

1. **S. Farhan**, S. Rahman, S. Aabira, G. Datta, S. Sultana, M.T. Amin, "High Speed and Low Power 8-Bit 12-Function ALU Using Transmission Gate Logic in 90nm Technology," 2025 Int. Conf. on Quantum Photonics, Artificial Intelligence and Networking (QPAIN).
2. **S. Farhan**, T.B. Jafar, M.T. Amin, "Design and Performance Analysis of Power-Efficient Compact BCD to Seven Segment Decoder Using Pass Transistor Logic," 2025 IEEE Int. Women in Engineering (WIE) Conference.
3. **S. Farhan**, P. Das, S. Shabab, "IoT-Based Health Monitoring System for Real-Time Vital Signs and Hospital Room Conditions Tracking," 2025 Int. Conf. on Quantum Photonics, Artificial Intelligence and Networking (QPAIN).
4. S. Rahman, **S. Farhan**, S. Aabira, K.N. Faisal, "GCAF-Net: An Attention-Enhanced Deep CNN Framework with Grad-CAM for Classifying Rare and Endangered Medicinal Plants in Bangladesh," 2025 Int. Conf. on Quantum Photonics, Artificial Intelligence and Networking (QPAIN).
5. S. Saha, Z.Z. Sarah, **S. Farhan**, M.T. Amin, "A Multi-Metric Comparative Analysis of CMOS Dynamic Latch Comparators Under Voltage Scaling," 2025 IEEE Int. Women in Engineering (WIE) Conference.
6. M.M. Haque, M.M. Rashid, **S. Farhan**, M.T. Amin, "Comparative Analysis of Configurable Logic Block Using Static CMOS, Transmission Gate and Gate Diffusion Input Techniques," 2025 28th Int. Conf. on Computer and Information Technology (ICCIT).
7. S. Rahman, I. Mohtasim, **S. Farhan**, K.N. Faisal, "Leveraging Power Spectrum-based Features and Machine Learning for Resource-Efficient Cough Classification in Respiratory Disease Monitoring," 2025 IEEE Int. Conf. on Signal Processing, Information, Communication and Systems (SPICSCON).

8. S. Rahman, S. Aabira, **S. Farhan**, S. Sultana, “Neural Network Enhanced PID Control of Automatic Voltage Regulators: Comparative Study with Classical and Regression Approaches,” 2025 IEEE Int. Conf. on Power, Electrical, Electronics and Industrial Applications (PEEIACON). (**Accepted**)
9. M.R. Kabbo, O.A. Ibrahim, M.M. Islam, **S. Farhan**, S. Sultana, M.T. Amin, “Design and Analysis of Hybrid Transmission Gate–MTCMOS 4:1 Multiplexer for Low-Power Nanoscale Circuits,” 2026 Int. Conf. on Power, Energy and Control for Industrial and Intelligent Systems (PECCII). (**Accepted**)

SELECTED PROJECTS

3D TCAD Optimization of a Triple Stacked Nanosheet GAAFET (Thesis) Silvaco TCAD

- Built and calibrated a 3D triple-stacked nanosheet GAAFET in Silvaco ATLAS.
- Optimized the device via gate-stack engineering, work-function tuning, inner-spacer design, and stack-configuration analysis for enhanced electrostatic control and drive performance.
- Simulated and analyzed short-channel effects (DIBL), electrostatic integrity, and temperature stability to assess device reliability.
- Extracted small-signal parameters (transconductance, output conductance, gate capacitance) and benchmarked performance against reported state of the art GAAFET architectures.

Modeling of an AlGaIn/GaN MIS-HEMT Wide Bandgap Transistor Silvaco TCAD

- Designed and simulated a 2D AlGaIn/GaN MIS-HEMT, a wide bandgap transistor suited to high power and high frequency communication systems.
- Modeled the AlGaIn/GaN heterostructure with polarization-induced two-dimensional electron gas (2DEG) formation and a metal-insulator-semiconductor gate stack.

Design of a Configurable Logic Block for FPGA Applications Cadence Virtuoso

- Designed and verified a reconfigurable FPGA logic block in Cadence using LUTs and flip-flops to realize arbitrary combinational and sequential logic.

Performance Analysis of a GaAs DG JLMOSFET Using TCAD Simulation Silvaco TCAD

- Analyzed the electrical characteristics of GaAs DG-JLMOSFET by studying transfer and output curves under varying gate oxide materials and thicknesses.

Design and Optimization of D-Shaped PCF-Based SPR Biosensor (Ongoing) COMSOL

- Designing a highly sensitive D-shaped photonic crystal fiber (PCF) surface plasmon resonance (SPR) biosensor in COMSOL Multiphysics, optimizing structural parameters for sensitivity.

TECHNICAL SKILLS

Device Simulation & TCAD: Silvaco ATLAS TCAD

IC & Circuit Design: Cadence Virtuoso, Microwind, DSCH2, PSpice, Proteus

HDL Design & Verification: Verilog HDL, ModelSim, Quartus Prime, EDA Playground

Programming & Data Analysis: Python, MATLAB, C, Assembly, ML/DL

Simulation Tools: COMSOL Multiphysics, Simulink, Proteus

Embedded Systems: Arduino, ESP32

Documentation & Productivity: LaTeX, Microsoft Office (Excel, PowerPoint, Word)

AWARDS & RECOGNITION

- **2023–2026 | MIST Dean’s List** — Awarded for consistent academic performance; maintained MIST Dean’s List for four consecutive years, and ranked 1st in the Electronics major.
- **2025 | Best Paper Award** — IEEE SPICSCON 2025; selected from 714 peer-reviewed submissions.
- **2025 | National Semiconductor Finalist** — Selected as one of 27 finalists at BEAR Summit & National Semiconductor Symposium 2025; presented “High-Speed and Low-Power 8-Bit 12-Function ALU Using Transmission Gate Logic in 90nm Technology.”
- **2025 | 2nd Runner-up** — Biotech Poster Showcase, BEAR Summit & National Semiconductor Symposium 2025.

EXTRACURRICULAR ACTIVITIES & LEADERSHIP

- Instructor — RTL Design Foundations with Verilog Bootcamp** 2026
IEEE EDS MIST Student Branch Chapter & MIST Robotics Club
Designed and delivered a six-session Verilog HDL bootcamp. Prepared lectures and hands-on labs using EDA Playground, ModelSim, and Quartus Prime.
- Department Representative (DR)** 2025–2026
Electrical and Electronic Engineering (EEE) Department, MIST
Represented EEE students across all levels in academic coordination and departmental activities.
- President** 2025–2026
MIST Robotics Club
Led planning, budgeting, and execution of the national robotics fest “Robolution” and organized student technical workshops.
- Vice Chair (Founding Member)** 2025–2026
IEEE Electron Devices Society (EDS) MIST Student Branch Chapter
Co-founded the chapter and promoted semiconductor device research and technical discussions.
- General Secretary** 2025–2026
IEEE MIST Student Branch
Coordinated IEEE activities, technical events, and student research initiatives.
- Campus Ambassador**
Mindsparks 2024 (AUST), Idea Innovation 5.0 (Interactive Cares), Anweshan 1.0 (SPE, RUET), and Energy Fest 1.0 (KUET).

INDUSTRIAL TRAINING

Completed short industrial attachments at two semiconductor design houses and a leading electronics manufacturer in Bangladesh.

Ulkasemi Pvt. Limited	1 Week May 2025
Neural Semiconductor Limited	1 Week May 2025
Walton Hi-Tech Industries Ltd.	1 Week May 2025

REFERENCES

Dr. Kazi Newaj Faisal Associate Professor Department of EEE, MIST Email: newajfaisal@gmail.com Phone: +880 1769-009270	Sunjida Sultana Assistant Professor Department of EEE, MIST Email: sunjida.eece@gmail.com Phone: +880 1919-909181 (Thesis Supervisor)
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